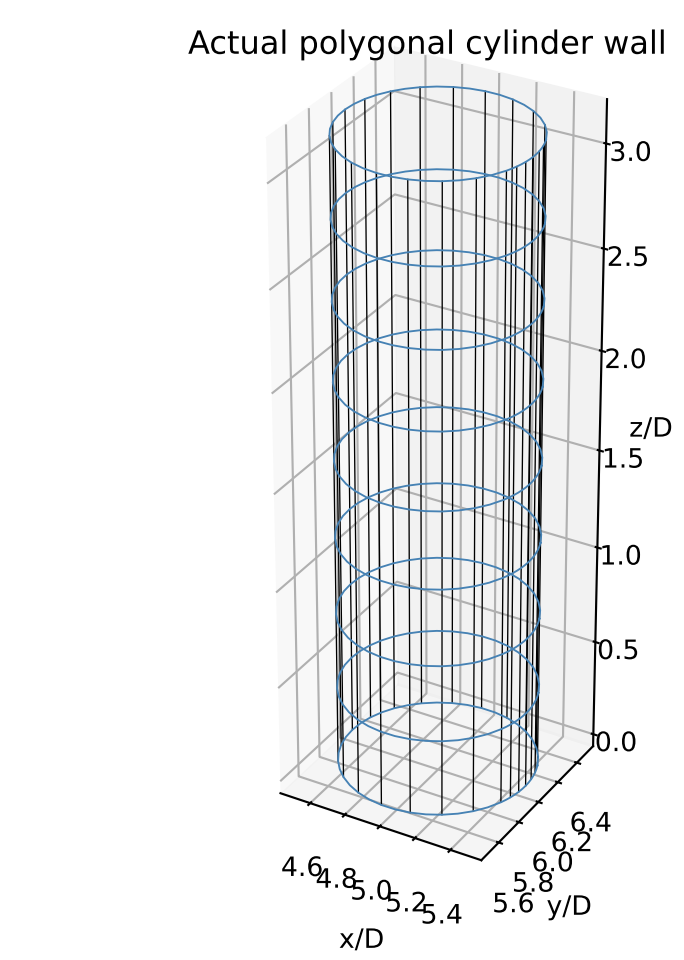
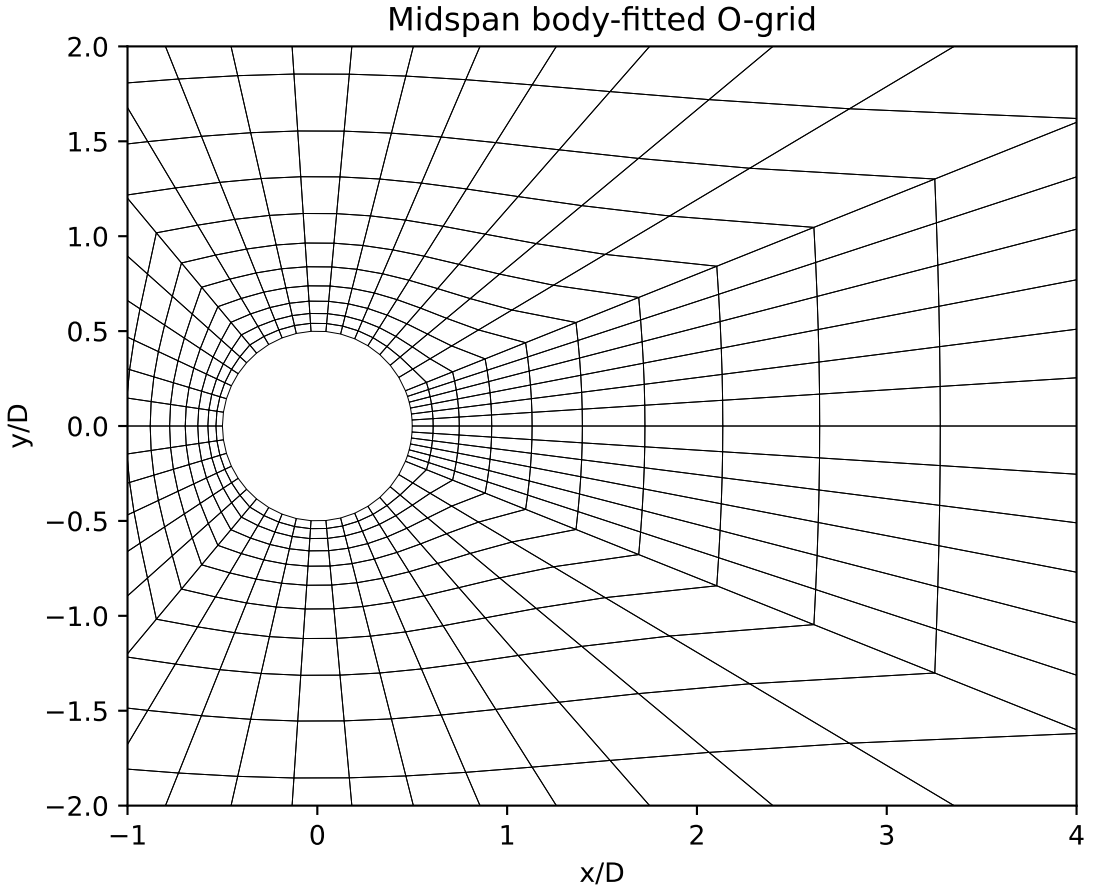
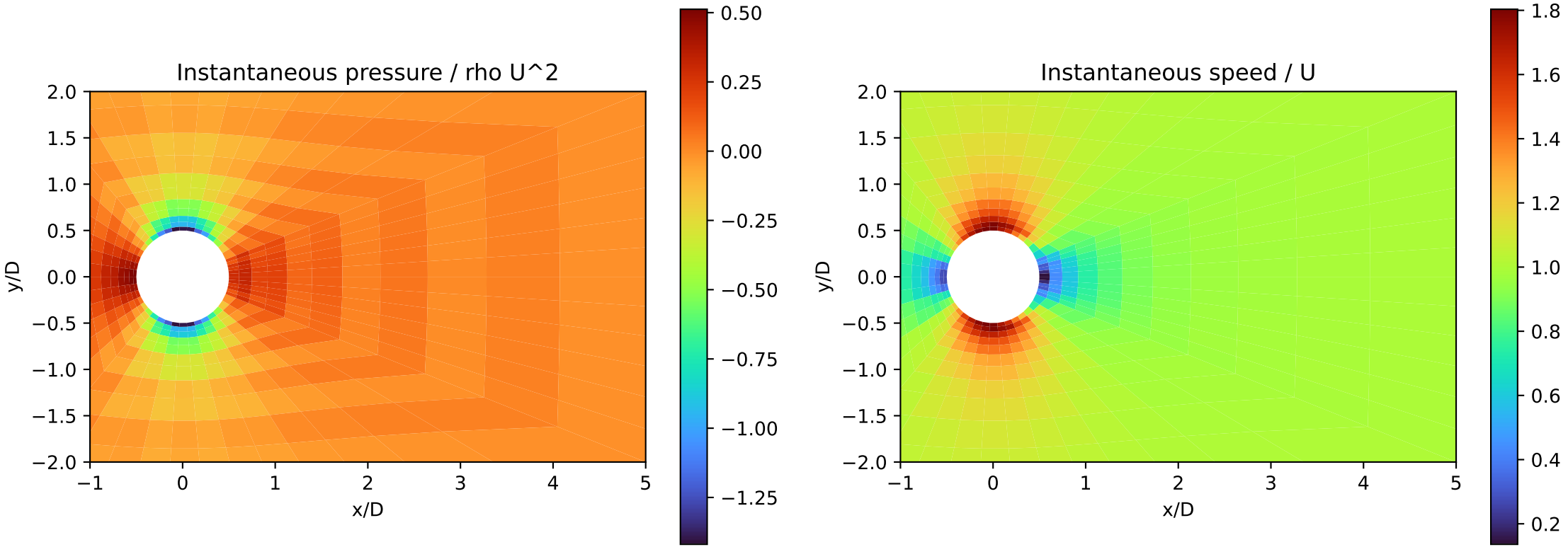
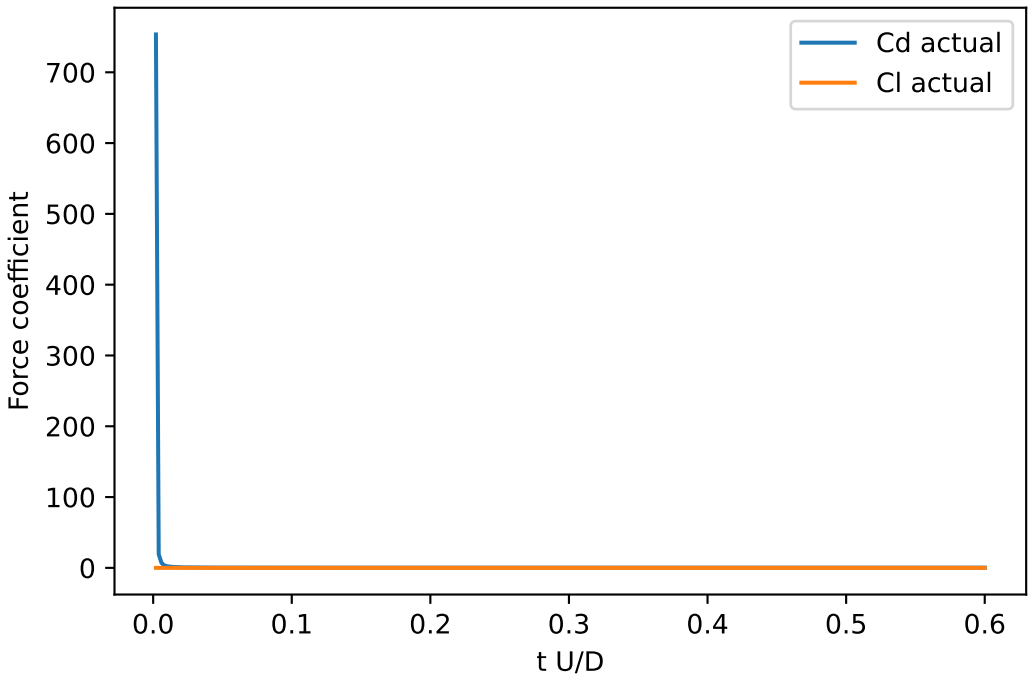
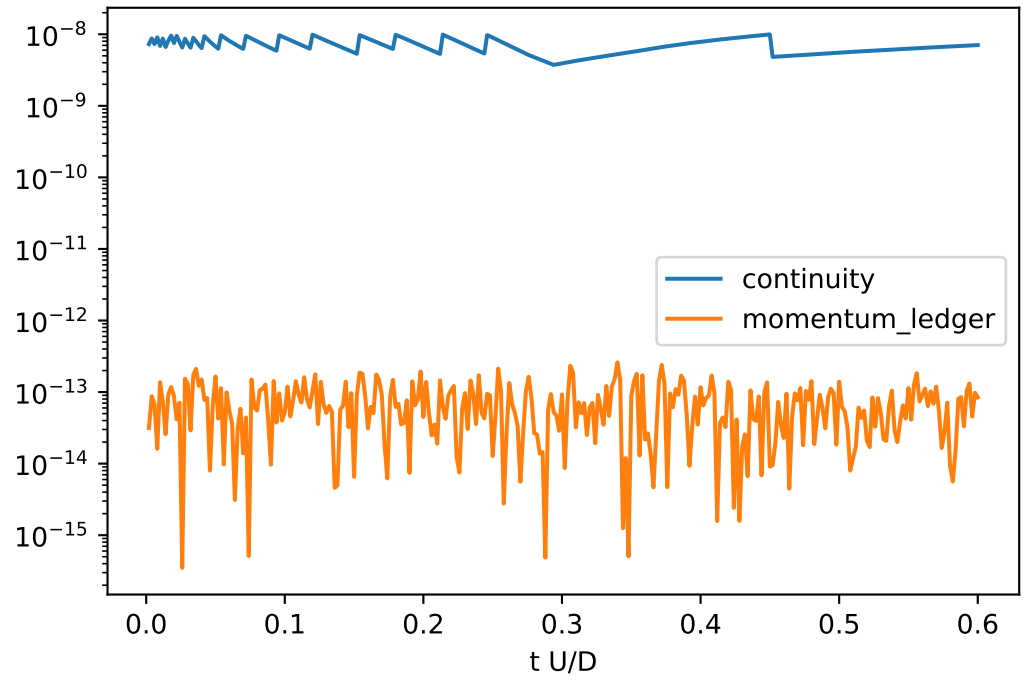
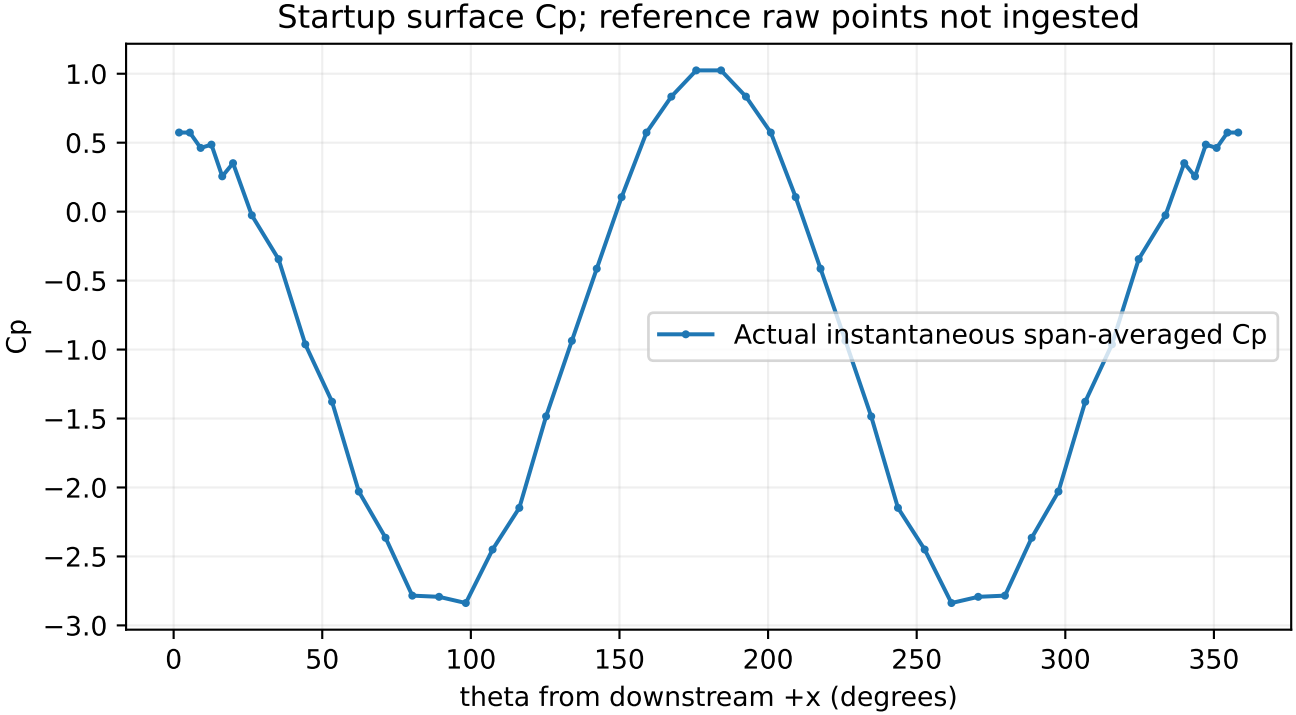

Actual polygonal cylinder wall
3.0
2.5
2.0
z/D
1.5
1.0
0.5
0.0
4.6
4.8
5.0
5.2
5.4
x/D
5.6
5.8
6.0
6.2
6.4
y/D


Midspan body-fitted O-grid
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
y/D
−1
0
1
2
3
4
x/D


Startup fields: not statistically converged LES
Instantaneous pressure / rho U^2
Instantaneous speed / U
y/D
x/D
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1
0
1
2
3
4
5
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2


Startup force, not time-averaged reference
Cd actual
Cl actual
Force coefficient
700
600
500
400
300
200
100
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
t U/D


Independent mass and momentum ledgers
10^{-8}
10^{-9}
10^{-10}
10^{-11}
10^{-12}
10^{-13}
10^{-14}
10^{-15}
continuity
momentum_ledger
0.0
0.1
0.2
0.3
0.4
0.5
0.6
t U/D


Startup surface Cp; reference raw points not ingested
Cp
theta from downstream +x (degrees)
Actual instantaneous span-averaged Cp
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
0
50
100
150
200
250
300
350